# Gammapy Analysis of archival VERITAS observations of Cassiopeia A SNR

Nicholas Bo You Chin*, David Kieda* for the VERITAS Collaboration†
Department of Physics & Astronomy, University of Utah, Salt Lake City, Utah*
veritas.sao.arizona.edu †

## Gamma-Ray Astronomy observes and studies the highest-energy form of electromagnetic radiation emitted by celestial objects and phenomena. It allows us to investigate extreme environments and energetic events, including supernova remnants, pulsars, blazars, black holes, neutron stars, galaxy clusters, and diffuse gamma-ray emission throughout the Milky Way.[2]

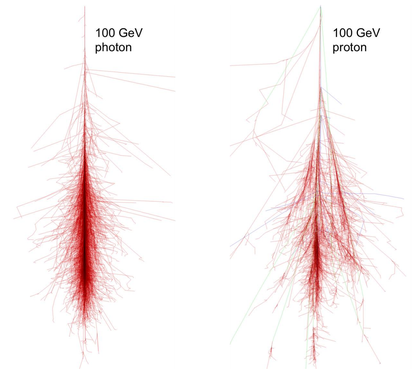



Figure 1. Monte Carlo Simulations of Photon and Protons induced showers.[3]



Image of the four telescopes at VERITAS
Credit: Center for Astrophysics | Harvard & Smithsonian

## VERITAS

(Very Energetic Radiation Imaging Telescope Array System) is a ground-based gamma-ray observatory with four 12 meter wide optical reflectors for GeV-TeV photon energy range gamma-ray astronomy.[4] It uses Imaging Atmospheric Cherenkov Telescope technique which allows determination of the gamma-ray induced extensive air shower core position and the precise reconstruction of the shower geometry.

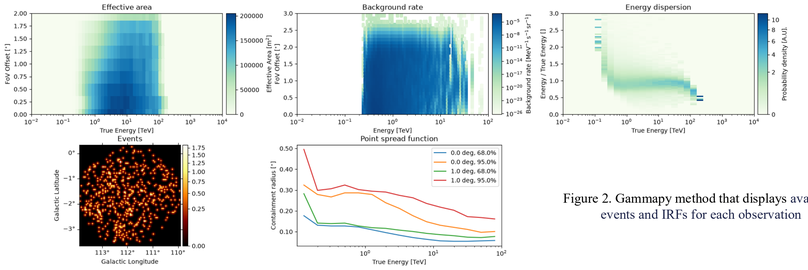



Figure 2. Gammapy method that displays available events and IRFs for each observation

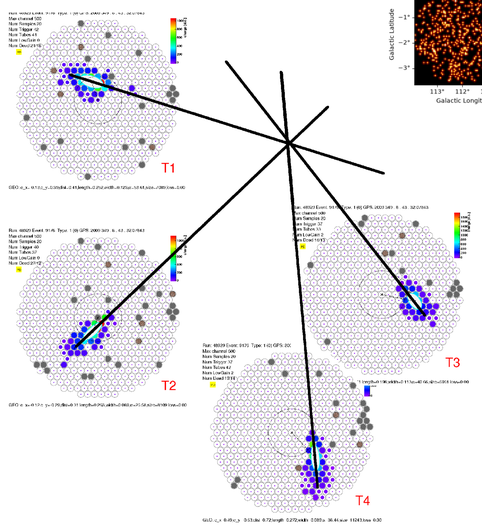



Figure 3. Gamma-ray Event Reconstruction

A **Python** package for **gamma-ray** astronomy

**Gammapy** is an open-source Python package for gamma-ray astronomy built on `numpy`, `scipy` and `astropy`. Gamma-ray observatories have traditionally relied on proprietary data and instrument-specific software. Next generation open facilities such as Cherenkov Telescope Array Observatory (CTAO) will make data and analysis tools more broadly accessible.[5] `gammapy` is therefore expanding use cases in the analysis of instruments including Fermi-LAT, H.E.S.S., MAGIC, VERITAS, and HAWC.[6]

## Cassiopeia A (Cas A) is one of the youngest supernova remnants (SNR) in the Milky Way. It is one of the brightest radio sources. Its optical emission is dominated by dense ejecta heated by reverse shock and several fast-moving knots.[7] The supernova event produces substantial amounts of heavy elements, motivating extensive studies in the X-ray regime.[8] Studies in the GeV-TeV emission suggest contributions from both inverse Compton scattering and neutral pion decay.[7,9,10] The evolutionary stage of Cas A also makes it a valuable probe for testing theoretical models that predict intrinsically asymmetric core-collapse supernova explosions.[11]



Figure 4. Cassiopeia A by Chandra X-ray Observatory and James Webb Space Telescope
Credit: NASA, ESA, CSA, STScI, D. Milisavljevic, I. De Looze, and T. Temim (2023)

## Results

We analyzed a subset of the complete dataset. In this analysis, we use over 20 hours of VERITAS observations of Cassiopeia A, primarily collected between September and November 2007. A three-dimensional (3D) analysis was performed to reproduce the results reported by Kumar et al.[6] Our analysis yielded a detection significance exceeding $6\sigma$. `gammapy` provides novel ways to quantify the expected contribution of background events, including "Field-of-View" (FOV) methods in which the background is estimated using a combination of smoothed matched runs from the same or an independent sample. Ongoing work seeks to quantify the unique systematic features present in this method that depend on the many factors that go into determining the proper matching of samples and the transformation between data and the background template.

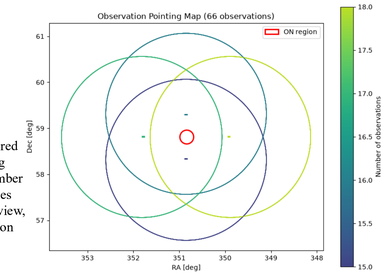



Figure 5. Spatial distribution of the observations used in this analysis. Colored markers indicate observation pointing centers, with colors representing the number of observations at each pointing. Circles indicate the corresponding 3.5° field-of-view, while the red circle marks the ON region centered on Cas A.

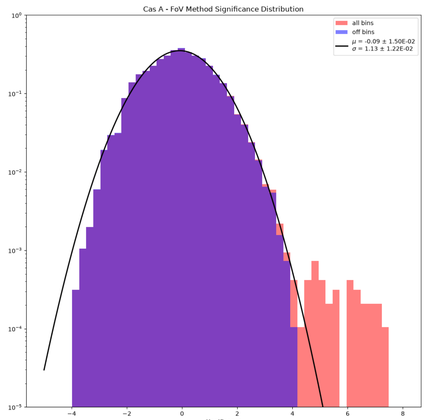
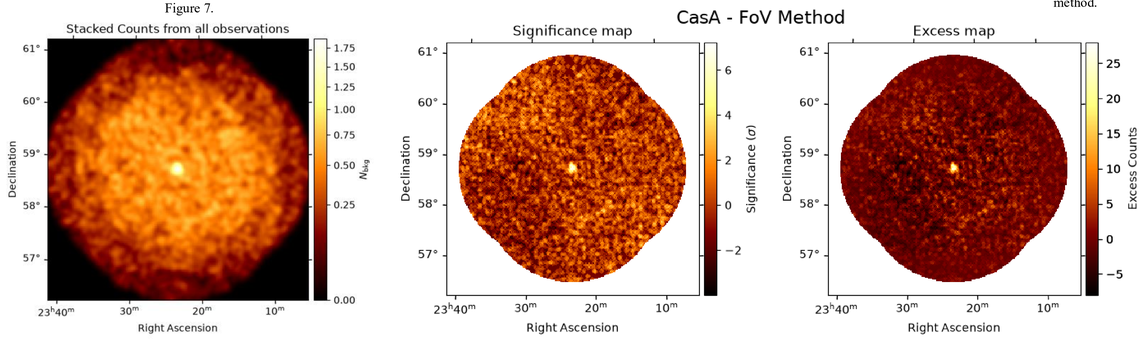



Figure 6. (Left) Significance distribution of all spatial bins (red) and OFF-region bins (blue) using the Field-of-View (FoV) background method. A Gaussian fit to the OFF-region distribution is shown in black, approximately consistent with a standard normal distribution. The positive tail extending beyond $6\sigma$ indicates bins with significant gamma-ray excess.

Figure 8. (Below) Significance and excess maps of Cassiopeia A. The left panel shows the statistical significance of gamma-ray emission, expressed in standard deviations $\sigma$, while the right panel shows the corresponding excess gamma-ray counts above the estimated background. Both maps were generated using the FoVbackground estimation method.

Figure 7.




**Acknowledgement**

This research is supported by grants from the U.S. Department of Energy Office of Science, the U.S. National Science Foundation and the Smithsonian Institution, and by NSERC in Canada. This research used resources provided by the Open Science Grid, which is supported by the National Science Foundation and the U.S. Department of Energy's Office of Science, and resources of the National Energy Research Scientific Computing Center (NERSC), a U.S. Department of Energy Office of Science User Facility operated under Contract No. DE-AC0205CH11231. We acknowledge the excellent work of the technical support staff at the Fred Lawrence Whipple Observatory and at the collaborating institutions in the construction and operation of VERITAS

**References**

[1] J. Devin, "The PeV Frontier: Status of Gamma-ray Astronomy after Two Decades with H.E.S.S., MAGIC, VERITAS and the New Window Recently Opened by HAWC and LHAASO," SF2A-2024: Proceedings of the Annual Meeting of the French Society of Astronomy and Astrophysics, p. 233, 2024.
[2] Wikipedia contributors, "Gamma-ray astronomy," *Wikipedia*, 2026. https://en.wikipedia.org/wiki/Gamma-ray_astronomy
[3] F. Schmidt. Images produced using the CORSIKA package. https://www-zeuthen.desy.de/~jknapp/fs/showerimages.html
[4] Wikipedia contributors, "VERITAS," *Wikipedia*. https://en.wikipedia.org/wiki/VERITAS
[5] A. Donath et al., "Gammapy: A Python package for gamma-ray astronomy," *Astron. Astrophys.*, vol. 678, Art. A157, 2023. https://doi.org/10.1051/0004-6361/202346488
[6] Gammapy Collaboration, "Gammapy: A Python package for gamma-ray astronomy." https://gammapy.org/
[7] A. U. Abeysekara et al., "Evidence for Proton Acceleration up to TeV Energies Based on VERITAS and Fermi-LAT Observations of the Cas A SNR," The Astrophysical Journal, 2020. https://doi.org/10.3847/1538-4357/ab8310
[8] Hwang, U., et al., "A Million-Second Chandra View of Cassiopeia A," The Astrophysical Journal Letters, vol. 615, pp. L117–L120, 2004. https://doi.org/10.1086/426186
[9] Yuan, Y., et al., "Fermi-LAT Detection of a Break in the Gamma-Ray Spectrum of the Supernova Remnant Cassiopeia A," The Astrophysical Journal, vol. 779, no. 2, article 117, 2013. https://doi.org/10.1088/0004-637X/779/2/117
[10] Saha, L., et al., "Origin of Gamma-Ray Emission in the Shell of Cassiopeia A," Astronomy & Astrophysics, vol. 563, article A88, 2014. https://doi.org/10.1051/0004-6361/201323218
[11] Hwang, U., and Laming, J. M., "A Chandra X-ray Survey of Ejecta in the Cassiopeia A Supernova Remnant," The Astrophysical Journal, vol. 746, no. 2, article 130, 2012. https://doi.org/10.1088/0004-637X/746/2/130